

An Introduction to Jacobian-Based Shape Servoing

Data-driven estimation and control for model-free deformable-object
manipulation

Lecture notes by Ignacio Cuiral-Zueco, with accompanying Python lab session

Abstract

Shape servoing is the task of bringing a deformable object to a desired target shape by means of robot manipulation. When no mechanical model of the object is available, the control law is organised around an *interaction Jacobian*: the local linear map relating gripper motion to the change it produces in the object's shape, as measured by a visual sensor such as an RGB-D camera. This Jacobian can be estimated directly from data and then inverted to define the shape controller. These notes develop the method through a single running example, a planar setup in which two 2D robots act on an elastic rod. We define the actuation and shape feature spaces, examine how the object's shape can be represented in several ways, explain how the Jacobian is estimated, and show how its pseudo-inverse closes the control loop. We then analyse the three characteristic ways in which a Jacobian-based shape control loop fails: loss of controllability, loss of observability, and mechanical instability. Each mathematical concept is accompanied by a short "intuition" box, and each section closes with a Python exercise from the accompanying lab session, so that a complete data-driven shape-servoing framework is built up incrementally.

Contents

1	Setup	1
2	Shape representation	3
3	Offline Jacobian estimation and its conditioning	5
3.1	The local linear map	5
3.2	Offline estimation: regression from exploration data	6
3.3	Span, rank, and conditioning	7
4	Closing the loop and the online update	8
4.1	Closing the loop	8
4.2	Online update: the Broyden rule	8
4.3	Three conditions for good closed-loop control	9
5	Three ways the loop fails	11
6	Summary	15

1 Setup

Two planar arms hold an elastic rod by its endpoints. Each gripper has three independent degrees of freedom (DoF): two translation DoFs and one orientation DoF. The current robot configuration can be represented through its end-effector (EE) pose vector

$$\mathbf{r} = [x_L, y_L, \theta_L, x_R, y_R, \theta_R]^\top \in \mathbb{R}^m, \quad m = 6,$$

where the subscripts L and R denote the left and right grippers, (x, y) is the gripper position in the world frame, and θ denotes the EE orientation (its rotation axis is perpendicular to the plane in which the setup is defined).

The rod is assumed to behave quasistatically, which means its physical state $\mathbf{x}$, a continuous curve with infinitely many degrees of freedom, evolves only when the robots actuate the rod through *small* changes in their configuration $\mathbf{r}$. The continuous state $\mathbf{x}$ cannot be measured directly from visual sensors, so shape servoing works instead with a reduced representation of $\mathbf{x}$: a finite feature vector $\mathbf{s} \in \mathbb{R}^n$. In our 2D rod example, the controller’s actuation space is the six-dimensional end-effector pose vector $\mathbf{r}$, and the feature space is the ten-dimensional vector $\mathbf{s}$ of sampled rod coordinates. We sample $P = 5$ points along the rod (yellow dots in Figure 1) and stack the planar coordinates (x_i, y_i) of the i -th sampled point into the feature vector

$$\mathbf{s} = [x_1, y_1, x_2, y_2, \dots, x_P, y_P]^\top \in \mathbb{R}^n, \quad n = 2P = 10 \quad (P = 5).$$

The control problem is to find the changes in EE pose $\mathbf{r}$ (i.e., the sequence of $\Delta\mathbf{r}$) that bring $\mathbf{s}$ as close as possible to the target value $\mathbf{s}^*$ that encodes the target shape.

One choice among many. The pair $(\mathbf{r}, \mathbf{s})$ used here is one simple example of an *actuation space* and a *feature space*. Other definitions are possible: a 3D actuation with three translation and three rotation DoF, or a feature vector $\mathbf{s}$ built from edge vectors, curvatures, contour descriptors, or visual feature descriptors rather than 2D point coordinates. We adopt the pair above for clarity; §2 discusses alternative choices of $\mathbf{s}$ and their properties.

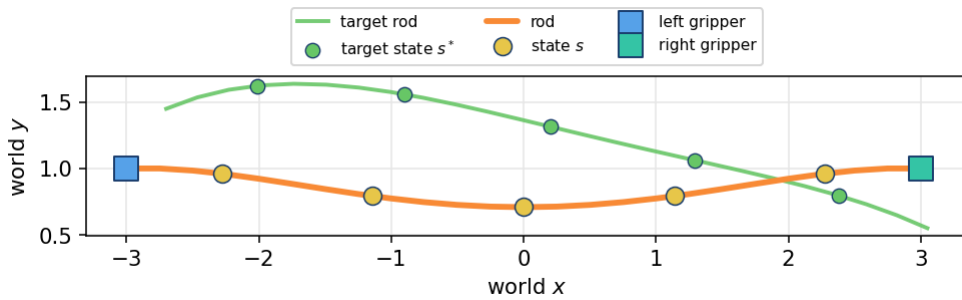


Figure 1: The control problem. Two grippers (blue, green squares) clamp an elastic rod (orange). Five points sampled along the rod (yellow) form the measured feature state $\mathbf{s}$. The target shape (green) and its sampled state $\mathbf{s}^*$ define where we want the rod to be; the task is to bring $\mathbf{s}$ to $\mathbf{s}^*$ by means of gripper actions.

We assume no knowledge of the physical model of the rod: no bending stiffness, no mass, and no boundary conditions. Everything the model-free controller needs is the local response of the measured features $\mathbf{s}$ (geometric data) with respect to changes in the robot configuration $\mathbf{r}$.

The shape servoing system runs in two phases, shown in Figures 2 and 3. In the *offline* phase, before any control takes place, the robots execute an exploration sequence (small end-effector

moves applied one degree of freedom at a time), and the measured responses are fit by least squares into an initial Jacobian estimate $\hat{\mathbf{J}}^{(0)}$. In the *online* phase, the controller compares the measured features $\mathbf{s}$ with the target $\mathbf{s}^*$, and turns the feature error $\mathbf{e}$ into robot actions $\Delta\mathbf{r}$ through the inverse (pseudoinverse) of the Jacobian matrix. Such actions produce a new rod state $\mathbf{x}$, and the perception block maps $\mathbf{x}$ to the new features $\mathbf{s}$. A Broyden estimator (to be introduced) refines $\hat{\mathbf{J}}$ online using the action values $\Delta\mathbf{r}$ and their resulting feature response $\Delta\mathbf{s}$. Each block is detailed in the following sections: §2 feature extraction, §3 offline Jacobian estimation and its conditioning, and §4 closing the loop and the online Broyden update.

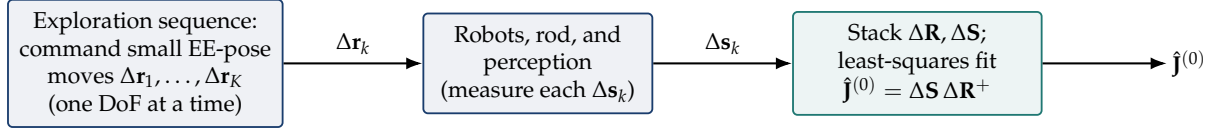


Figure 2: Offline initialisation. The robots run an exploration sequence, moving one end-effector degree of freedom at a time; each commanded $\Delta\mathbf{r}_k$ and its measured feature response $\Delta\mathbf{s}_k$ are stacked into matrices $\Delta\mathbf{R}, \Delta\mathbf{S}$ and fit by least squares to obtain the initial Jacobian estimate $\hat{\mathbf{J}}^{(0)}$, which is later used on the online control loop.

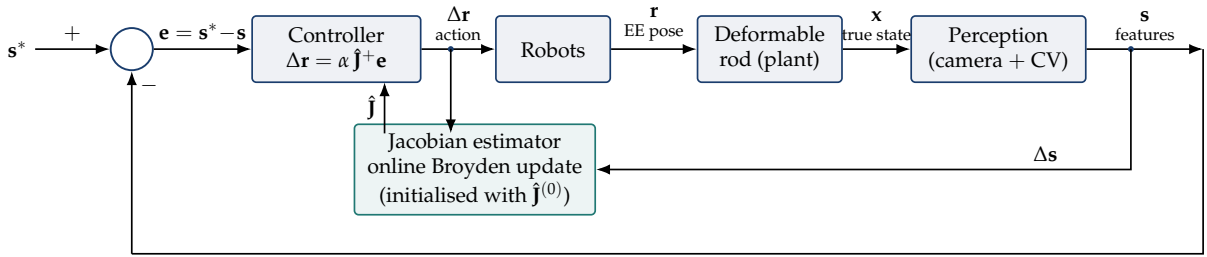


Figure 3: Online control loop. The forward path (blue) turns the shape error $\mathbf{e} = \mathbf{s}^* - \mathbf{s}$ into robot actions $\Delta\mathbf{r}$; the rod responds and settles at the new true state $\mathbf{x}$, which the perception block maps to the measured features $\mathbf{s}$. The feedback loop defines the next error value. The online estimator (green) refines the interaction Jacobian $\hat{\mathbf{J}}$ using the past action value $\Delta\mathbf{r}$ and its corresponding feature response $\Delta\mathbf{s}$. Such online estimation process is initialised with the offline estimate $\hat{\mathbf{J}}^{(0)}$.

Lab exercise | Exercise 1

Define two different end-effector pose vectors, a nominal (initial) pose $\mathbf{r}$ and a target pose $\mathbf{r}^*$. For each, solve the rod equilibrium and sample it to obtain the corresponding feature vectors $\mathbf{s}$ and $\mathbf{s}^*$. Compute the shape error $\mathbf{e} = \mathbf{s}^* - \mathbf{s}$ and its norm $\|\mathbf{e}\|$. Plot the current rod, its sampled features, the grippers, and the target. (See code/ex1_setup.py.)

2 Shape representation

In §1 we used 2D point coordinates to define $\mathbf{s}$. That was one choice among several reasonable ones, and the choice of feature space matters: it determines which shape changes the controller can *perceive*, and therefore which shape changes the Jacobian can describe.

Defining $\mathbf{s}$ and $\dot{\mathbf{s}}$

As introduced earlier, a shape representation is a smooth map from the rod's continuous physical configuration $\mathbf{x}$ to a finite feature vector $\mathbf{s} \in \mathbb{R}^n$. Its time derivative,

$$\dot{\mathbf{s}} = \frac{d\mathbf{s}}{dt},$$

is the velocity in feature space.

Physical state vs. measured features. The rod has a real, continuous shape: a curve with infinitely many points. The controller has access only to the finite vector $\mathbf{s}$ extracted from that curve through sensors (e.g., a camera) and processing (e.g., point sampling or feature detection). Different rod curves can produce the same $\mathbf{s}$, and the same curve can produce different values of $\mathbf{s}$ depending on the chosen shape representation.

We now introduce three common choices of features to illustrate how they carry different information and represent shapes differently.

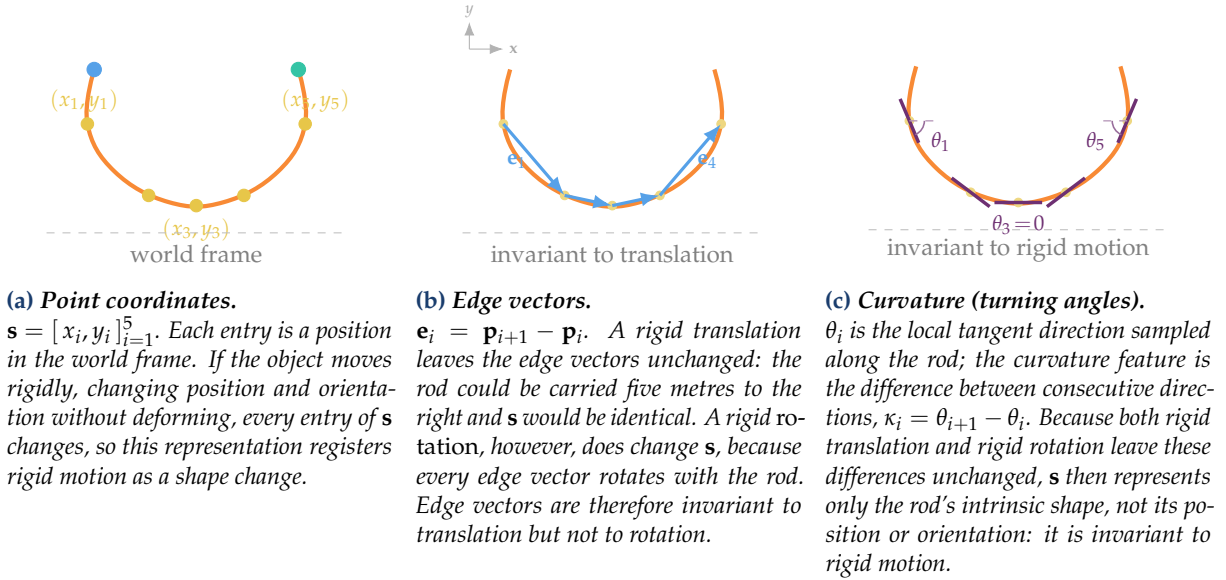


Figure 4: Three ways to represent the same U-shape with a finite feature vector $\mathbf{s}$. Each choice leads to different characteristics regarding invariance to rigid translations or rotations.

What changes from one representation to the next

The three representations carry progressively less information about where the rod *is* with respect to the world frame, and progressively more about its *intrinsic* shape:

- **Point coordinates.** A pure translation of the rod by 0.20 [m] shifts every x_i by 0.20, so the measured change in error is large. This is the representation used in our 2D rod example.
- **Edge vectors.** Each entry is $\mathbf{e}_i = \mathbf{p}_{i+1} - \mathbf{p}_i$. A pure translation now leaves $\mathbf{s}$ unchanged; a rigid rotation does not, because the edge vectors rotate with the rod.
- **Curvature (turning angles).** Each entry is the turning angle between consecutive edges, $\kappa_i = \theta_{i+1} - \theta_i$, where θ_i is the direction of edge $\mathbf{e}_i$; the P sampled points give $P - 1$ edge

directions and $P - 2$ turning angles. Both rigid translation and rigid rotation leave $\mathbf{s}$ unchanged; only an actual change in shape affects $\mathbf{s}$.

The U-shape test

Consider a rod with a U-shape that moves rigidly, i.e., it is translated and rotated without undergoing material deformation. Sampling the rod at five points to define $\mathbf{s}$ as point coordinates makes the feature state change substantially, whereas using the sampled curvature values as $\mathbf{s}$ leaves it unchanged. Figure 5 illustrates this.

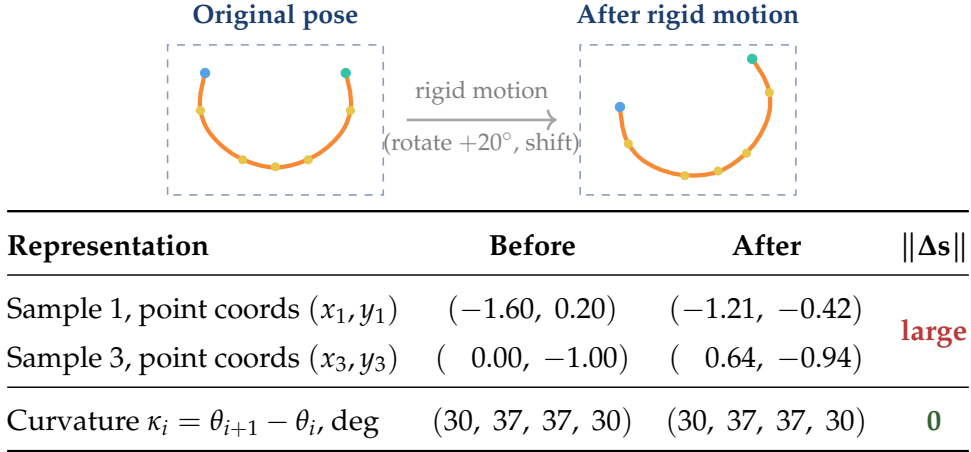


Figure 5: The same physical U-shape, before and after a rigid motion (translation and rotation). Point-coordinate features change substantially; curvature features are identical. Using point coordinates reflects a change in the state; using curvature values does not.

No choice of $\mathbf{s}$ is universally correct; it depends on the objective. For example, use point coordinates when the position of the rod matters, or use a rigid-motion-invariant representation when only the shape matters. The choice of $\mathbf{s}$ also affects the Jacobian: a gripper motion that produces $\dot{\mathbf{s}} \neq 0$ under one representation may produce $\dot{\mathbf{s}} = 0$ under another. This relates directly to the observability discussed in §5.

Finally, representations can be combined. For instance, $\mathbf{s}$ may stack both point coordinates and curvatures,

$$\mathbf{s} = [x_1, y_1, \dots, x_P, y_P, \theta_2 - \theta_1, \dots, \theta_P - \theta_{P-1}]^\top.$$

Lab exercise | Exercise 2

Implement the three feature types (point coordinates, edge vectors, and curvature) as functions of the rod's sampled points. Apply a rigid transformation to a U-shape, first a pure translation and then a translation combined with a rotation, and for each feature map compute the change $\|\Delta \mathbf{s}\|$ it induces. Confirm the results are coherent with the analysis from Figure 5. (See code/ex2_representation.py.)

3 Offline Jacobian estimation and its conditioning

3.1 The local linear map

The controller's goal is to bring $\mathbf{s}$ to $\mathbf{s}^*$. Without a model of the rod, we approximate how changes in robot configuration generate changes in feature space $\mathbf{s}$ through the local map

$\mathbf{J} : \dot{\mathbf{r}} \mapsto \dot{\mathbf{s}}$, where $\mathbf{J}$ is the *interaction Jacobian* $\mathbf{J}(\mathbf{r}) \in \mathbb{R}^{n \times m}$ (note that in our rod example $n = 10$, $m = 6$):

$$\dot{\mathbf{s}} = \mathbf{J}(\mathbf{r}) \dot{\mathbf{r}}. \quad (1)$$

The mechanics that set $\mathbf{J}$ are unknown to the controller, so $\mathbf{J}$ cannot be written down from a model: it must be estimated from data. **From now on $\mathbf{J}$ denotes the *true* interaction Jacobian, which the controller cannot observe, and $\hat{\mathbf{J}}$ its estimate.**

Intuition | What a Jacobian is

A Jacobian is the matrix counterpart of the *slope* of a scalar function. For a scalar $f(x)$, the slope df/dx describes how the output changes when the input changes by a small amount. For a vector-valued f , the slope becomes a matrix whose entry J_{ij} describes how output i changes when input j changes. Here J_{ij} answers the following question: *if gripper DoF j of $\mathbf{r}$ moves by a small amount, say 0.01 m, by how much does sampled coordinate i of $\mathbf{s}$ shift?* Column j is therefore the shape change, in feature space $\mathbf{s}$, produced by actuating DoF j alone. Figure 6 shows $\mathbf{J}$ as a heat-map of these columns.

Interaction Jacobian $\mathbf{J}$ (one column per gripper DoF)

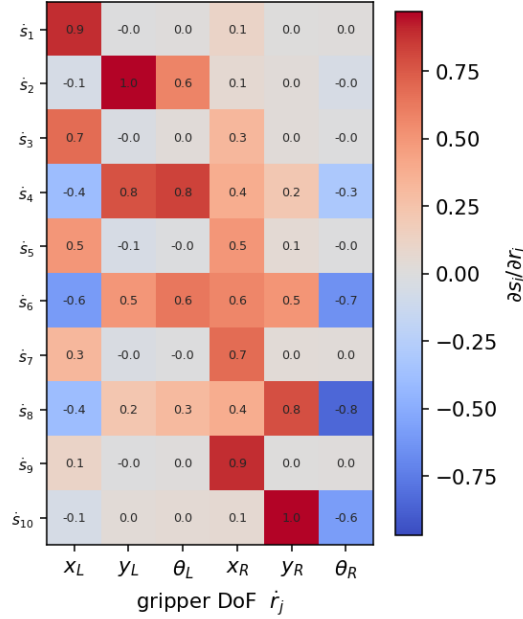


Figure 6: The interaction Jacobian as a heat-map. Each cell (i, j) is $\partial s_i / \partial r_j$, that is, the influence of gripper DoF j on feature i .

3.2 Offline estimation: regression from exploration data

Before any closed-loop control begins, the controller commands a sequence of small exploration motions and records the rod's response. We move each gripper DoF independently at four amplitudes, giving 24 measured pairs in total. Each pair is $(\Delta \mathbf{r}_k, \Delta \mathbf{s}_k)$ with

$$\Delta \mathbf{r}_k = \mathbf{r}_k - \mathbf{r}_0, \quad \Delta \mathbf{s}_k = \mathbf{s}_k - \mathbf{s}_0.$$

Stacking these pairs column-wise gives the data matrices

$$\Delta R = [\Delta r_1, \dots, \Delta r_{24}] \in \mathbb{R}^{6 \times 24}, \quad \Delta S = [\Delta s_1, \dots, \Delta s_{24}] \in \mathbb{R}^{10 \times 24}.$$

The initial estimate is the linear map that best explains $\Delta S \approx \hat{J} \Delta R$ in a least-squares sense:

$$\hat{J}^{(0)} = \arg \min_{\hat{J}} \|\Delta S - \hat{J} \Delta R\|_F^2 = \Delta S \Delta R^+, \quad (2)$$

where $\Delta R^+ = \Delta R^\top (\Delta R \Delta R^\top)^{-1}$ is the right Moore–Penrose pseudo-inverse of the wide exploration matrix ΔR (6×24), well defined as long as the exploration motions span all six DoF, which we guarantee by actuating each DoF separately. The result is a 10×6 matrix like the one in Figure 6, with one column per gripper DoF.

Intuition | Why this regression works

Each exploration motion is one measurement: we apply a step $\Delta \mathbf{r}_k$ and record the response $\Delta \mathbf{s}_k$. If the rod were exactly linear we would have $\Delta \mathbf{s}_k = \mathbf{J} \Delta \mathbf{r}_k$ for one common $\mathbf{J}$ at every step. The rod is only approximately linear over the exploration amplitudes, so the 24 equations cannot all hold exactly; we ask instead for the single $\hat{\mathbf{J}}$ that minimises the total squared mismatch. Equation (2) is the closed-form solution to that least-squares problem.

3.3 Span, rank, and conditioning

For control we will invert $\hat{\mathbf{J}}$, so it must have full column rank and be well conditioned. We characterise this through the span, rank, and singular values of the Jacobian (read off the estimate $\hat{\mathbf{J}}^{(0)}$ in the lab), in the sense made precise by the following definitions.

Intuition | Span: which shape changes the grippers can reach

The column span of $\mathbf{J}$ is the set of all $\dot{\mathbf{s}}$ that some choice of $\dot{\mathbf{r}}$ can produce. Each column is the shape change from perturbing one gripper DoF; together the six columns span at most a 6-dimensional subspace of the 10-dimensional shape space. Shape changes inside the span can be commanded; those outside it cannot. Control towards the target requires the column span of $\mathbf{J}$ to contain the deformation direction the controller needs, i.e. the direction proportional to $\mathbf{s}^* - \mathbf{s}$.

Intuition | Rank: the number of genuinely independent input directions

The rank of $\mathbf{J}$ is the dimension of its column span. A 10×6 Jacobian has rank at most 6; full column rank ($\text{rank}(\mathbf{J}) = 6$) means each gripper DoF contributes a shape change the others cannot reproduce. When the rank drops, there exist actions $\dot{\mathbf{r}} \neq 0$ with $\dot{\mathbf{s}} = \mathbf{J} \dot{\mathbf{r}} = 0$: the grippers move, but the measured shape does not.

A finer analysis comes from the singular value decomposition (SVD):

$$\mathbf{J} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top, \quad \mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_6), \quad \sigma_1 \geq \dots \geq \sigma_6 \geq 0. \quad (3)$$

Intuition | Singular values σ_i : the gains of independent channels

The SVD reorganises $\mathbf{J}$ into six independent channels. Each maps a particular combination of gripper DoF (a column of $\mathbf{V}$) to a particular shape change (a column of $\mathbf{U}$), with σ_i as the scalar gain on that channel.

Scalar example. With a single input and a single output (a 1×1 Jacobian), say $J = 2.5$, the only singular value is $\sigma_1 = 2.5$: an input $\dot{r} = 0.01$ gives an output 0.025. To invert, $J^+ = 1/\sigma_1 = 0.4$. If instead $\sigma_1 = 0.001$, producing the same output needs an input of 25, an amplification by a factor of 1000. At $\sigma_1 = 0$ the output is unreachable and the inverse does not exist. The matrix case behaves the same way along each channel: every σ_i must be bounded away from zero for $\mathbf{J}^+$ to be well conditioned.

The singular values yield two summary quantities:

$$w = \sqrt{\det(\mathbf{J}^\top \mathbf{J})} = \prod_i \sigma_i, \quad \kappa = \sigma_{\max}/\sigma_{\min}. \quad (4)$$

Intuition | Manipulability w and condition number κ

w is the product of all singular values, equal to the volume of the ellipsoid of shape changes reachable by a small all-DoF perturbation. If any $\sigma_i \rightarrow 0$ the ellipsoid collapses and $w \rightarrow 0$. Here w measures the controller's overall capacity to actuate $\mathbf{s}$ in independent directions: **larger w is better, and $w = 0$ means at least one direction of $\mathbf{s}$ can no longer be actuated.**

κ is the ratio of the easiest to the hardest direction to actuate (largest over smallest σ_i). $\kappa = 1$ means every direction responds equally; $\kappa \rightarrow \infty$ as $\sigma_{\min} \rightarrow 0$. **Smaller κ is better; a very large κ signals a direction that produces almost no change in $\mathbf{s}$.** §5 tracks $\sigma_{\min}$, w , and κ together as they collapse near a singularity.

Lab exercise | Exercise 3

Estimate the interaction Jacobian $\hat{\mathbf{J}}^{(0)}$ at the nominal EE pose by offline regression: explore each DoF, gather the input/output changes $\Delta\mathbf{R}, \Delta\mathbf{S}$, and fit $\hat{\mathbf{J}}^{(0)} = \Delta\mathbf{S} \Delta\mathbf{R}^+$. Read its rank, manipulability w , and condition number κ from the SVD. (See `code/ex3_offline_jacobian.py`.)

4 Closing the loop and the online update

With a well-conditioned estimate $\hat{\mathbf{J}}$ in hand (§3), the controller inverts it to drive the shape, and refreshes it online as the rod deforms away from where it was measured.

4.1 Closing the loop

The controller closes the loop by inverting the estimate. A clean inverse $\hat{\mathbf{J}}^{-1}$ exists only for a square, full-rank matrix. Here $\hat{\mathbf{J}} \in \mathbb{R}^{n \times m}$ with $n = 10 > m = 6$: it is rectangular (more shape features than actuation DoF) and therefore not invertible in the usual sense. We use the Moore–Penrose pseudo-inverse $\hat{\mathbf{J}}^+$:

$$\Delta \mathbf{r} = \alpha \hat{\mathbf{J}}^+ (\mathbf{s}^* - \mathbf{s}), \quad \hat{\mathbf{J}}^+ = (\hat{\mathbf{J}}^\top \hat{\mathbf{J}})^{-1} \hat{\mathbf{J}}^\top. \quad (5)$$

Intuition | The pseudo-inverse $\hat{\mathbf{J}}^+$

For a tall, full-column-rank $\hat{\mathbf{J}}$, the Moore–Penrose pseudo-inverse $\hat{\mathbf{J}}^+ = (\hat{\mathbf{J}}^\top \hat{\mathbf{J}})^{-1} \hat{\mathbf{J}}^\top$ is the best linear substitute for an inverse: given an error $\mathbf{e} = \mathbf{s}^* - \mathbf{s}$, the step $\hat{\mathbf{J}}^+ \mathbf{e}$ is the $\Delta \mathbf{r}$ whose predicted shape change $\hat{\mathbf{J}} \Delta \mathbf{r}$ is closest to $\mathbf{e}$ in the least-squares sense. In SVD coordinates $\hat{\mathbf{J}}^+$ acts on channel i with gain $1/\sigma_i$. This is why a small σ_i in $\hat{\mathbf{J}}$ becomes a large gain in $\hat{\mathbf{J}}^+$: the controller commands large gripper motions precisely along the directions in which the rod responds least.

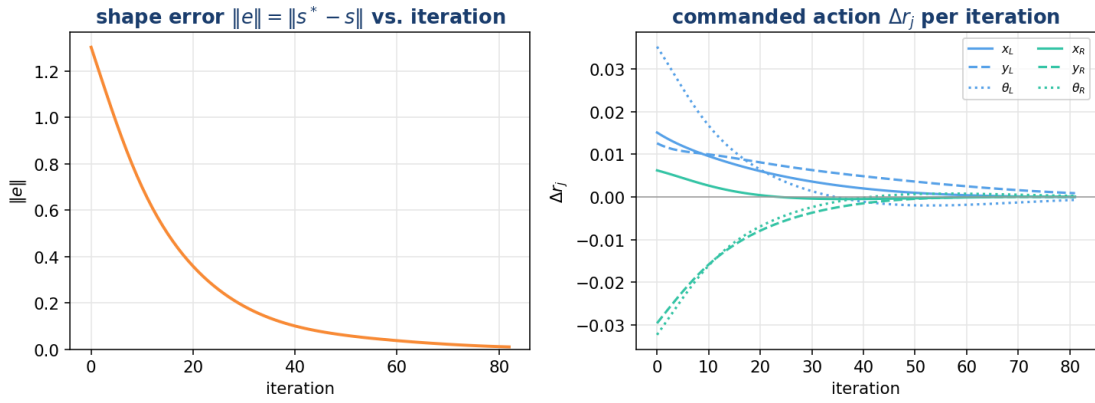


Figure 7: The closed loop in operation, with a small gain $\alpha = 0.05$. Left: the shape error $\|\mathbf{e}\|$ falls smoothly and steadily towards zero as the pseudo-inverse step $\Delta \mathbf{r} = \alpha \hat{\mathbf{J}}^+ \mathbf{e}$ is applied and the rod is re-measured. Right: the commanded action $\Delta \mathbf{r}_j$ at each iteration, which shrinks to zero as the error vanishes.

4.2 Online update: the Broyden rule

The offline estimate is local: it is accurate near the initial configuration $\mathbf{r}_0$ where it was measured, but as the control loop runs and the rod deforms, the true $\mathbf{J}(\mathbf{r})$ changes and $\hat{\mathbf{J}}^{(0)}$ loses accuracy. To keep $\hat{\mathbf{J}}$ current, the controller updates it at every iteration using the data the loop already produces.

At iteration k the robot commanded a step $\Delta \mathbf{r}$ and observed a shape change $\Delta \mathbf{s}$. If $\hat{\mathbf{J}}$ were exact we would have $\Delta \mathbf{s} = \hat{\mathbf{J}} \Delta \mathbf{r}$; in general there is a residual $\Delta \mathbf{s} - \hat{\mathbf{J}} \Delta \mathbf{r} \neq 0$ quantifying what the estimate failed to predict. The *Broyden update* corrects $\hat{\mathbf{J}}$ to absorb that residual:

$$\hat{\mathbf{J}}^{(k+1)} = \hat{\mathbf{J}}^{(k)} + \frac{(\Delta \mathbf{s} - \hat{\mathbf{J}}^{(k)} \Delta \mathbf{r}) \Delta \mathbf{r}^\top}{\Delta \mathbf{r}^\top \Delta \mathbf{r}}, \quad (6)$$

with $\Delta \mathbf{r} = \mathbf{r}^{(k+1)} - \mathbf{r}^{(k)}$ and $\Delta \mathbf{s} = \mathbf{s}^{(k+1)} - \mathbf{s}^{(k)}$.

Intuition | What Broyden is doing

The numerator is the prediction error: what the rod actually did, minus what $\hat{\mathbf{J}}$ predicted. The outer product with $\Delta \mathbf{r}^\top$ projects that error back along the direction just moved, and

the denominator $\|\Delta \mathbf{r}\|^2$ normalises by the size of the step. Adding this rank-one correction yields the updated $\hat{\mathbf{J}}^{(k+1)}$ used at the next step.

The update is the multivariable generalisation of a secant slope: with no analytical model of the rod, we refine the estimate using the data measured during control. The sensitive term is the denominator. When $\|\Delta \mathbf{r}\|$ is small but $\|\Delta \mathbf{s}\|$ is large, the factor $1/\|\Delta \mathbf{r}\|^2$ makes the correction arbitrarily large. §5 (Category 3) returns to this: at a mechanical instability such as buckling, a tiny $\Delta \mathbf{r}$ produces a large $\Delta \mathbf{s}$, so Eq. (6) injects a large spurious correction into $\hat{\mathbf{J}}$, and the next pseudo-inverse commands no longer reflect the rod's true response.

4.3 Three conditions for good closed-loop control

With $\hat{\mathbf{J}}$ in the loop, three conditions must hold at once. Each one also governs the offline exploration and the online update, so any of them can fail at the start of the task or appear partway through it:

1. **Controllability / reachability.** The columns of the true $\mathbf{J}$ span the shape changes needed to reach $\mathbf{s}^*$, and the estimate $\hat{\mathbf{J}}$ has not collapsed its own column span, so the inverse map $\hat{\mathbf{J}}^+$ remains well defined.
2. **Observability.** The chosen features $\mathbf{s}$ register the rod's physical evolution, so the measured $\Delta \mathbf{s}$ feeding Eq. (6) is informative.
3. **Linearity.** The model $\dot{\mathbf{s}} = \mathbf{J}\dot{\mathbf{r}}$ remains a good local approximation, and $\Delta \mathbf{s}$ comes from that smooth regime rather than from a discontinuous response.

Each of the next three categories corresponds to one of these conditions failing.

Lab exercise | Exercise 4

Starting from the offline estimate $\hat{\mathbf{J}}^{(0)}$ of Exercise 3, add the Broyden update of Eq. (6) and complete the closed control loop that drives $\mathbf{s}$ to the target. With the Broyden update on the loop converges quickly; with a fixed estimate it converges much more slowly. (See `code/ex4_control_loop.py`.)

5 Three ways the loop fails

Category 1 | Controllability singularities

Analytical side. The grippers move, but the resulting shape change is negligible. Some direction of gripper motion leaves the physical system essentially unchanged, so the columns of $\mathbf{J}$ along that direction shrink (DoF $\dot{\mathbf{r}}_j$ loses influence on the rod): $\sigma_{\min}(\mathbf{J}) \rightarrow 0$, $w \rightarrow 0$, $\kappa \rightarrow \infty$. When the controller asks $\mathbf{J}^+$ to correct an error along that direction, the gain $1/\sigma_{\min}$ grows without bound: the commanded gripper step becomes large while the shape changes very little.

Broyden update. The estimate receives $\Delta \mathbf{r} \neq 0$ but $\Delta \mathbf{s} \approx 0$. The residual reduces to $-\hat{\mathbf{J}} \Delta \mathbf{r}$, so the update shrinks $\hat{\mathbf{J}}$'s column along $\Delta \mathbf{r}$. After a few iterations $\hat{\mathbf{J}}$ also reports that direction as inactive, which is precisely the direction the controller still needs in order to move the shape.

Where it arises.

- *Taut state.* The rod is pulled straight; outward gripper motion produces no $\dot{\mathbf{s}}$, only tension. Figure 8 increases the separation through this regime and shows $\sigma_{\min}(\mathbf{J})$ collapsing.
- *Slack state.* The grippers are closer than the rod's resting length; gripper motion is taken up by the slack rather than transmitted through the rod.
- *Rigid-body and symmetry redundancies.* Several gripper motions produce the same $\dot{\mathbf{s}}$, so $\mathbf{J}$ is genuinely rank-deficient on its input side.

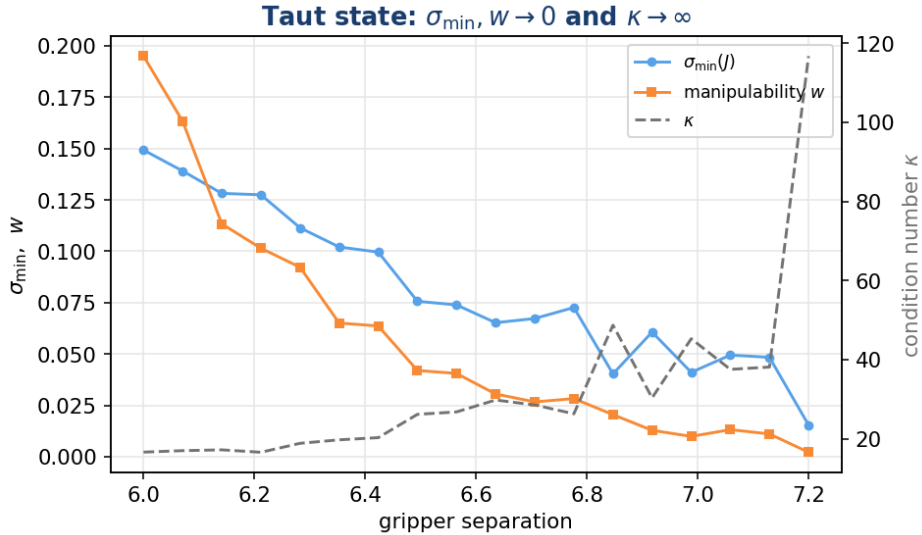


Figure 8: Taut state (Category 1). As the gripper separation grows, the rod is pulled into tension: $\sigma_{\min}(\mathbf{J})$ and the manipulability w collapse while the condition number κ rises from about 17 to 117. The pseudo-inverse would amplify commands as $1/\sigma_{\min}$ to keep the shape moving.

Category 2 | Observability singularities

Analytical side. The rod *is* deforming in response to gripper motion, but the chosen features $\mathbf{s}$ are insensitive to that particular deformation. Now the rows of $\mathbf{J}$, not the columns, are the problem: $\dot{\mathbf{s}} \approx 0$ because of the features, so the controller receives an uninformative $\Delta \mathbf{s} \approx 0$ even though the rod actually moved.

The link to §2 is direct: this is what happens when the representation is too coarse, too symmetric, or too invariant to register the change.

Broyden update. The numerical signature is the same as Category 1 (a non-zero $\Delta \mathbf{r}$ paired with $\Delta \mathbf{s} \approx 0$), but the cause is the feature map rather than the physics. The relevant rows of $\hat{\mathbf{J}}$ decay toward zero, so the controller stops attributing any feature response to those gripper directions.

Where it arises.

- *Sub-parametrised features.* Twisting a smooth, untextured rod about its axis: the material twists but the sampled centreline does not move.
- *Symmetric features.* Move one gripper up by the same amount the other moves down. The rod clearly deforms, but a scalar feature such as the midpoint height stays at zero. Figure 9 shows this case.
- *Self-occlusion.* A folded garment hides part of itself; rows of $\mathbf{s}$ vanish mid-task.

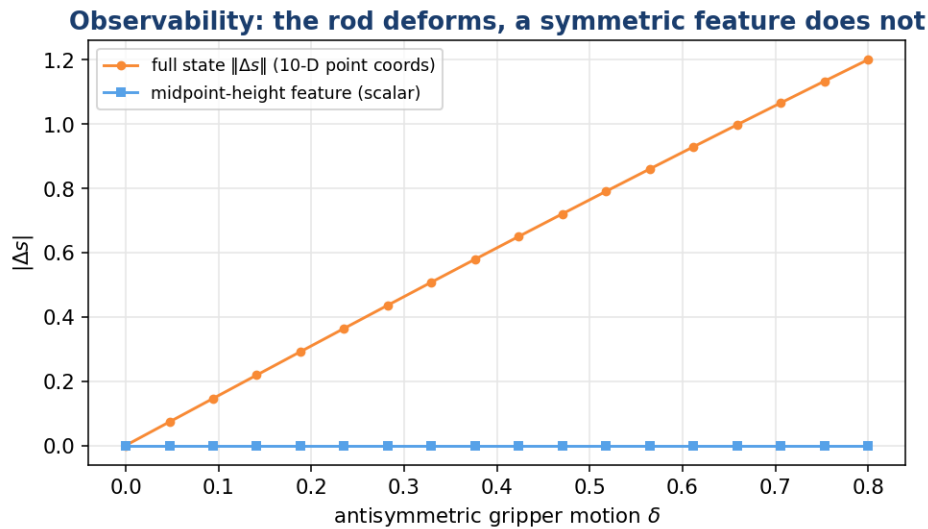


Figure 9: Observability (Category 2). An antisymmetric vertical motion (one gripper up, the other down) clearly deforms the rod, so the full 10-D point-coordinate state grows steadily; yet a single symmetric scalar feature (the midpoint height) stays at zero and does not register the deformation.

Category 3 | Mechanical instabilities

Analytical side. Categories 1 and 2 were both rank deficiencies of $\mathbf{J}$. This one is qualitatively different: the linear model itself ceases to hold. A small input $\dot{\mathbf{r}}$ produces a large, near-discontinuous $\dot{\mathbf{s}}$, and no finite matrix $\mathbf{J}$ can describe that relationship.

The mechanical reason is that the rod stores internal elastic energy as it strains, and at certain configurations the Hessian of the elastic potential loses positive definiteness:

$$\det(H_{\text{energy}}) = 0 \quad \Longleftrightarrow \quad \lambda_{\min}(H_{\text{energy}}) \rightarrow 0. \quad (7)$$

The current equilibrium ceases to be stable; the rod releases the stored elastic energy and settles into a different equilibrium shape on its own.

Broyden update. During the snap the gripper moved very little ($\|\Delta \mathbf{r}\|$ small) while the shape changed by a large amount ($\|\Delta \mathbf{s}\|$ large). In Eq. (6),

$$\hat{\mathbf{f}}^{(k+1)} - \hat{\mathbf{f}}^{(k)} = \frac{\overbrace{(\Delta \mathbf{s} - \hat{\mathbf{f}}^{(k)} \Delta \mathbf{r})}^{\text{very large}} \Delta \mathbf{r}^\top}{\underbrace{\|\Delta \mathbf{r}\|^2}_{\text{very small}}},$$

so the numerator is dominated by the large $\Delta \mathbf{s}$ and the denominator is the square of a tiny step. A single iteration injects a large spurious correction into $\hat{\mathbf{J}}$, and the next pseudo-inverse $\hat{\mathbf{J}}^+$ produces commands unrelated to the rod's actual response.

Where it arises.

- *Euler buckling.* Axial compression. The straight rod is stable until $\lambda_{\min}(\mathbf{H}_{\text{energy}})$ crosses zero, then snaps sideways.
- *Bi-stable snap-through.* A pre-curved shell pushed past an energy barrier snaps to a second equilibrium.
- *Supercoiling.* Twisting a rubber cord steadily: torque rises linearly until the rod buckles into a helical writhe.

Intuition | Energy Hessian and positive definiteness

A rod at rest sits at a local minimum of its elastic potential energy $U(\mathbf{x})$, where $\mathbf{x}$ is the rod's physical configuration. Near that minimum $\mathbf{U}$ looks like a bowl: every small perturbation increases the stored energy, so the rod is pushed back to rest. The Hessian $\mathbf{H}_{\text{energy}} = \partial^2 U / \partial \mathbf{x}^2$ describing this bowl has all positive eigenvalues: it is positive definite, curving upward in every direction (a ball placed inside returns to the bottom). When the smallest eigenvalue drops to zero, the bowl flattens in one direction, like a trough. Past that point the eigenvalue is negative and the bowl becomes a saddle: the equilibrium is unstable, and a small perturbation sends the rod to a different shape far away. That is a buckling event, and there the linear model $\dot{\mathbf{s}} = \mathbf{J}\dot{\mathbf{r}}$, which presumes a deep, stable bowl, no longer applies.

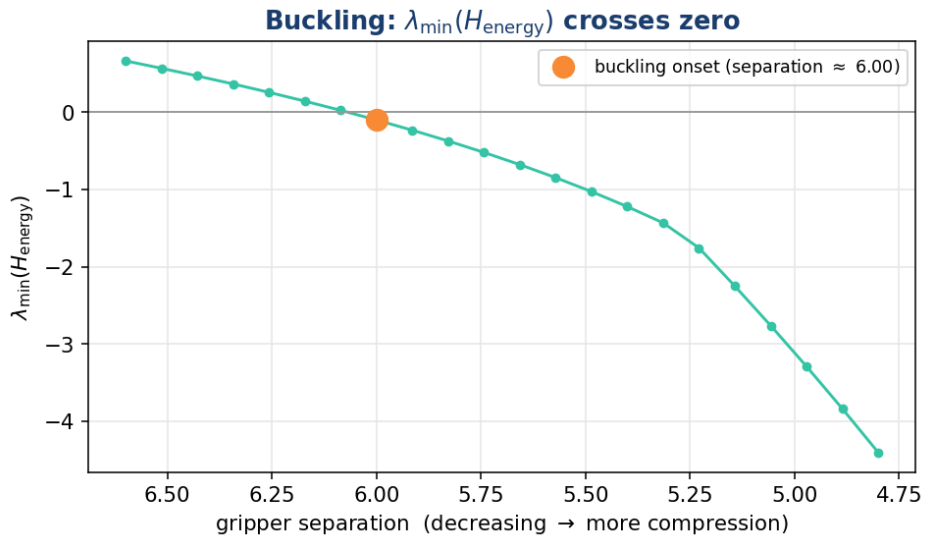


Figure 10: Buckling (Category 3). As the gripper separation decreases the rod is axially compressed. The smallest eigenvalue of the straight-rod energy Hessian crosses zero (marked); below that separation the straight equilibrium is unstable and the rod snaps sideways.

Lab exercise | Exercise 5

Reproduce all three failure modes with the same rod and Jacobian tools: increase the gripper separation and observe $\sigma_{\min}$, w , and κ collapse in the taut state (Figure 8); apply an antisymmetric motion to expose a feature insensitive to the deformation (Figure 9); and compress the rod until $\lambda_{\min}(\mathbf{H}_{\text{energy}})$ crosses zero (Figure 10). (See `code/ex5_failure_modes.py`.)

6 Summary

Category	What breaks	Analytical failure	Effect on $\hat{\mathbf{J}}$ (Broyden)
1. Controllability (taut, slack, redundancy)	Columns of $\mathbf{J}$	$\sigma_{\min}(\mathbf{J}) \rightarrow 0, w \rightarrow 0, \kappa \rightarrow \infty$. $\mathbf{J}^+$ amplifies error by $1/\sigma_{\min}$.	$\Delta \mathbf{r} \neq 0, \Delta \mathbf{s} \approx 0$. Update shrinks $\hat{\mathbf{J}}$'s column toward zero.
2. Observability (textureless or over-symmetric features)	Rows of $\mathbf{J}$	Rank drop on the output side. $\dot{\mathbf{s}} \approx 0$ even though the rod moves; no shape-change signal, $\Delta \mathbf{s} \approx 0$.	Same signature as Category 1, with the feature map as the cause. The relevant rows of $\hat{\mathbf{J}}$ decay.
3. Mechanical instabilities (buckling, snap-through, supercoiling)	The relation $\dot{\mathbf{s}} = \mathbf{J}\dot{\mathbf{r}}$ itself	$\det(\mathbf{H}_{\text{energy}}) = 0$. A small $\dot{\mathbf{r}}$ produces a large, autonomous $\dot{\mathbf{s}}$; no finite $\mathbf{J}$ describes it.	$\ \Delta \mathbf{r}\ $ small, $\ \Delta \mathbf{s}\ $ large. $1/\ \Delta \mathbf{r}\ ^2$ injects a large spurious correction into $\hat{\mathbf{J}}$ in one step.

A robust shape-control loop needs $\mathbf{J}$ well-conditioned (Category 1), features that actually register the deformation (Category 2, with §2 as the broader context), and the linear model to remain valid (Category 3). Because $\hat{\mathbf{J}}$ is updated online by Eq. (6), each failure has two effects: the immediate inversion is unreliable, and the estimate the next iteration relies on is degraded. The three categories map exactly onto the three preconditions of §4.2, and the lab of §5 reproduces each one in isolation.